



5.3 Integration and the trapezoidal rule

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The indefinite integral is the antiderivative: $\int x^n dx = x^{n+1}/(n+1) + c$ for $n \neq -1$. The definite integral $\int [a,b] f(x) dx$ represents the (signed) area under $y=f(x)$, evaluated as $F(b)-F(a)$ using the Fundamental Theorem of Calculus. When f is only known at discrete points or is difficult to integrate exactly, the trapezoidal rule gives an approximation: $\int [a,b] f(x) dx \approx (h/2)[y_0 + y_n + 2(y_1 + \dots + y_{n-1})]$, where $h=(b-a)/n$. It overestimates a concave-up (convex) function and underestimates a concave-down function.

Worked example: follow the reasoning

Estimate $\int [0,4] (x^2+1) dx$ using the trapezoidal rule with 4 intervals ($h=1$), then compare to the exact value.
Solution: $x=0,1,2,3,4$ give $y=1,2,5,10,17$. Trapezoidal: $(1/2)[1+17+2(2+5+10)]=(1/2)(52)=26$.
Exact: $[x^3/3+x]$ from 0 to 4 = $64/3+4 \approx 25.33$. The trapezoidal rule overestimates slightly, as expected for a concave-up function.

Quick check · multiple choice

The trapezoidal rule applied to a concave-up function typically gives an estimate of the definite integral that is:

- A Exactly correct
- B An underestimate
- C An overestimate
- D Undefined

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The rate of water flow into a tank (litres/min) was recorded at 5-minute intervals over 20 minutes: t (min): 0,5,10,15,20; rate: 12,18,25,30,28. (a) Use the trapezoidal rule with these data points to estimate the total volume of water that flowed into the tank. (b) The exact flow rate is later modelled by $r(t) = -0.03t^2 + 1.8t + 12$. Find the exact volume using integration, and compare it to your trapezoidal estimate. [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $h=5$: $(5/2)[12+28+2(18+25+30)]=2.5(40+146)=2.5(186)=465$ litres. (b)
 $\int[0,20](-0.03t^2+1.8t+12)dt=[-0.01t^3+0.9t^2+12t]$ from 0 to 20 = $(-80+360+240)=520$ litres.
The trapezoidal estimate (465 L) underestimates the exact value (520 L) by 55 L, consistent with the rate function being concave down ($r''(t)=-0.06<0$) throughout the interval.

Quick check answer

C. Straight-line chords over a concave-up (convex) curve lie above the curve, so the trapezoidal rule overestimates the integral.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

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